

Find the Mistake: Indexes, Ratios and Convergence

- Every problem quotes a value someone actually got. Your job is the correct value — and, where asked, the move that produced the wrong one.
 - Watch the starting index. A rule written for $n \geq 1$ means the first term is at $n = 1$, and a sum written from $k = 1$ has no $k = 0$ term in it.
 - An infinite geometric series has a sum only when the common ratio is between -1 and 1 . Check that before reaching for the formula.
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1. A sequence is given by $a_n = 4n + 3$ for $n \geq 1$. Asked for the fifth term, a student wrote 19.

(a) Find the correct fifth term.

Answer: _____

(b) Explain the error that produced 19.

Answer: _____

2. Asked to evaluate $\sum_{k=1}^4 (2k + 1)$, a student wrote 25. Find the correct value of the sum.

Answer: _____

3. A geometric sequence begins 5, 15, 45. A student reports the common ratio as 10, “because $15 - 5 = 10$ ”.

(a) Find the correct common ratio.

Answer: _____

(b) Using the correct ratio, find the sixth term of the sequence.

Answer: _____

4. Consider the expansion of $(x + 2)^4$, written in descending powers of x .

(a) Expand $(x + 2)^4$ completely.

Answer: _____

(b) A student reports the coefficient of the *third* term as 32. Find the correct coefficient of the third term.

Answer: _____

5. A student applies the infinite geometric sum formula to $3 + 6 + 12 + 24 + \cdots$ and writes $\frac{3}{1-2} = -3$.

(a) Explain why that value is meaningless for this series.

Answer: _____

(b) The series $3 + \frac{3}{2} + \frac{3}{4} + \cdots$ does have a sum. Find it.

Answer: _____

6. The arithmetic sequence $a_n = 5n - 2$ is defined for $n \geq 1$. Asked for the sum of its first 20 terms, a student wrote 1008, having added 21 terms. Find the correct sum.

Answer: _____

7. In the expansion of $(2x + 3)^5$, a student reports the coefficient of x^2 as 540, having used 2 where the term needs 2 squared. Find the correct coefficient of x^2 .

Answer: _____

8. A geometric sequence has all terms positive. Its third term is 20 and its fifth term is 80.

(a) Find the common ratio.

Answer: _____

(b) Find the first term.

Answer: _____

(c) Find the sum of the first six terms.

Answer: _____